

Tim S. Haubold

Curriculum Vitae

Education

- 2023 **PhD**, *Institute for applied mathematics, University of Hannover, Summa Cum Laude*, Thesis title: *Applications of Special Functions in High Order Finite Element Methods* (Advisor: Prof. Dr. Sven Beuchler)
Keywords: Numerical analysis, orthogonal polynomials, symbolic computation, high order finite elements
- 2017 **B.Sc and M.Sc**, *University of Bremen*, Master thesis title: *Hydrostatic reconstructed FORCE scheme and application to the non-unique Riemann solutions for the shallow water equations* (Advisor: Prof. Dr. Alfred Schmidt)
Keywords: Finite Volumes, shallow water equations

Employment

Current

- 2026 **Postdoctoral researcher**, *École nationale des ponts et chaussées / Inria Paris - Team SERENA with Alexandre Ern*
Keywords: Unfitted Finite Element, Time-dependent problem, Wave equation
- 2023–2026 **Postdoctoral researcher**, *Georg-August Universität Göttingen with Christoph Lehrenfeld as main collaborator*
Keywords: Unfitted Finite Elements, Maxwell's equation, Numerical Analysis

Past

- 2017–2023 **Predocctoral researcher**, *Gottfried-Wilhelm-Leibniz-Universität Hannover under supervision of Sven Beuchler*
Keywords: Symbolic Computation, High order finite element, Basis construction
- 2014–2017 **Student Assistant**, *Universität Bremen, SFB 747: Mikrokaltumformen*

Publications (Main author)

- 2024 **High order biorthogonal functions in $H(\text{Curl})$** , *Haubold, Beuchler, Schöberl*, SIAM Journal on Numerical Analysis, Volume 62, Nr. 5, Pages 2331–2348
<https://doi.org/10.1137/23M1606794>
- 2023 **Recursion Formulas for Integrated Products of Jacobi Polynomials**, *Haubold, Beuchler, Pillwein*, Constructive Approximation, Volume 59, pages 583–618
<https://doi.org/10.48550/arXiv.2105.08989>
- 2022 **Recurrences for Quadrilateral High-Order Finite Elements**, *Haubold, Beuchler, Pillwein*, Mathematics in Computer Science, Volume 16, article number 32
<https://doi.org/10.1007/s11786-022-00547-2>

Proceedings

- 2023 **Applications of Special Functions in High Order Finite Element Methods**, *Haubold, Beuchler, Pillwein, Schöberl*, Mini-Workshop: Multivariate Orthogonal Polynomials: New synergies with Numerical Analysis, Workshop Report 2023,43
10.14760/OWR-2023-43
- 2021 **Recursion relations for hp-FEM Element Matrices on quadrilaterals**, *Haubold, Beuchler, Pillwein*, PAMM, Volume 21, Issue 1, e202100200
<https://doi.org/10.1002/pamm.202100200>
- 2019 **Symbolic Evaluation of hp-FEM Element Matrices**, *Haubold, Beuchler, Pillwein*, PAMM, Volume 19, Issue 1, e201900446
<https://doi.org/10.1002/pamm.201900446>

Preprints

- 2026 **High order biorthogonal functions for the discrete de Rham complex**, *Beuchler, Haubold, Schöberl*, arXiv
<https://arxiv.org/pdf/2609.28345>

In preperation (Main author)

- 2026 **Parameter-robust unfitted finite element methods for a Maxwell interface problem**, *Haubold, Lehrenfeld*
- 2026 **Optimal decay rate of high order local operator**, *Haubold, Maier*
- 2026 **Unfitted discontinuous galerkin methods for the linear wave equation**, *Ern, Haubold*

Teaching

- 2026 **Exercises (English)**, *Éléments finis*
- 2025-2026 **Lecture and Exercise**, *Mathematisch orientiertes Programmieren*, Joint with Dr. Igor Voulis
- 2025 **Exercises**, *Functional Analysis*
- 2024-2025 **Lecture and Exercises**, *Numerics of partial differential equation III*, Joint with Prof. Dr. Christoph Lehrenfeld
- 2024 **Lecture**, *Mathematisch orientiertes Programmieren*, Joint with Dr. Igor Voulis
- 2023-2024 **Exercises**, *Numerics of partial differential equation I and II*
- 2017-2023 **Exercises**, *Numerics I, Numerics II, Numerics of partial differential equations (german and english), Numerical methods for continuum mechanics, Numerics for teachers, mathematics for special-needs teachers, mathematics for civil engineers, . . .*, in average 4 hours at the blackboard
- 2022 **Lecture and Exercise**, *Interaktive algorithmische Einführung in die Finiten Elemente*, Examiner: Prof. Dr. Sven Beuchler
- 2020 **External Lecture at Leibniz Fachhochschule Hannover**, *Statistics I*

Visits:

- 19.-23.11.18 **JKU Linz**, *Linz, Austria*, Working with Veronika Pillwein
14.-15.09.23 **TU Wien**, *Vienna, Austria*, Working with Joachim Schöberl
15.10.25 **TU Hamburg**, *Hamburg, Germany*, Invited seminar talk
09.-12.12.25 **KIT**, *Karlsruhe, Germany*, Working with Roland Maier
01.10.26 **LmB**, *Besançon, France*, Invited seminar talk

Talks:

- 2026 **Gamm Annual Meeting**, *Stuttgart, Germany*, Co-Chair of Section 18
2025 **International Conference on Domain Decomposition Methods**, *Milan, Italy*
2025 **Gamm Annual Meeting**, *Poznan, Poland*
2025 **SIAM CSE**, *Fort Worth, USA*
2024 **Chemnitz Finite Element Symposium**, *Chemnitz, Germany*
2024 **WCCM-PANACM**, *Vancouver, Canada*
2024 **Eccomas**, *Lisboa, Portugal*
2024 **Gamm Annual Meeting**, *Magdeburg, Germany*
2023 **Oberwolfach Mini-Workshop: Multivariate Orthogonal Polynomials: New synergies with Numerical Analysis**, *Oberwolfach, Germany*
2023 **Chemnitz Finite Element Symposium (on tour)**, *Leibnitz, Austria*
2022 **WAND**, *Salzburg, Austria*
2022 **Gamm Annual Meeting**, *Aachen, Germany*
2021 **ICOSAHOM**, *Vienna, Austria (online)*
2021 **Gamm Annual Meeting**, *Kassel (online)*
2019 **Chemnitz Finite Element Symposium (on tour)**, *Essen, Germany*
2019 **Gamm Annual Meeting**, *Vienna, Austria*
2019 **GAMM-Workshop 2019 - "Young Academics Workshop"**, *Sion, Switzerland*
2018 **Chemnitz Finite Element Symposium**, *Chemnitz, Germany*

Skills

Finite element software: NGSolve, ngxfem

Programming: Python, MATLAB, C++ (basic)

Computer algebra: Mathematica

Languages

German native speaker

English fluent

Additional Information

- 2026 **Co-Chair of Section 18 at Gamm Annual Meeting**
- 2024 **FE Stokes Re-Pair (a nerdy card game)**, *Haubold, Lehrenfeld, van Beeck, Zerbinati, Zienecker*
<https://fe-nerd-games.github.io/FES Stokes RePair/>
- 2022-2023 **Employee representative and boardmember of the Institute for applied mathematics, University of Hannover**
- 2020-2022 **Employee vice representative of the Institute for applied mathematics, University of Hannover**
- 2023 **Local organizer of the first internal Poster day at the Ifam, University of Hannover**
- 2022 **Part of the local organizer of annual Northern German Colloquium (NoKo) 2022**

References

Sven Beuchler, *PhD - Advisor*
Alexandre Ern, *Postdoctoral mentor*
Christoph Lehrenfeld, *Postdoctoral mentor*
Joachim Schöberl, *Collaborator*

Collaborators

Sven Beuchler, *Leibniz University Hannover, DE*
Veronika Pillwein, *Johannes Kepler University Linz, AT*
Joachim Schöberl, *TU Wien, AT*
Christoph Lehrenfeld, *University Göttingen, DE*
Roland Maier, *Karlsruher Institut für Technologie, DE*
Alexander Heinlein, *TU Delft, NL*
Alexandre Ern, *ENPC Paris / Inria Paris, FR*